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Break-Even Analysis

Exam code: 0450, 0986 & 0264, 0774

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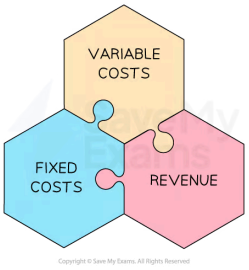
The concept of break-even

- **The break-even point is the number of units that need to be sold for total costs to equal the sales revenue**
  - It helps businesses understand the **minimum level of sales or output they need to achieve in order to cover all costs**
  - This helps business managers to make **informed decisions about pricing and production volumes**

Elements of a break-even analysis



Variable costs, fixed costs and sales revenue are all used in calculating the break-even point

- **Fixed costs** do not change, regardless of the level of production or sales
  - E.g. rent, salaries and insurance
- **Variable costs** vary with the level of production or sales
  - E.g. raw materials, direct labour costs, packaging and shipping costs
- **Sales revenue** is money gained from selling products, calculated using the formula

$$\text{Sales revenue} = \text{Number of items sold} \times \text{Selling price}$$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- **Break-even charts show the number of units a business must sell in order to break-even**
- In order to **construct a break-even chart,** the business needs to know the estimated fixed costs, variable costs and sales revenue

Worked Example

Steps to construct a break-even chart

Step 1: Identify fixed costs, variable costs and sales revenue at different levels of output

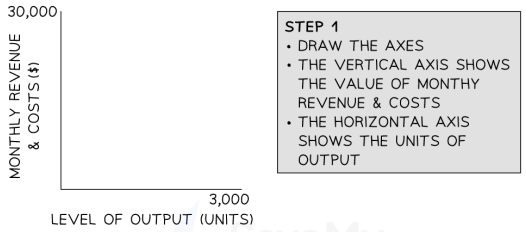
- Tee Crazy Ltd has the following cost and revenue data:
  - Fixed costs are $8,800 per year
  - The variable costs of each t-shirt is $4
  - Each t-shirt is sold for a price of $10
  - Its factory can produce a maximum output of 3000 t-shirts per year

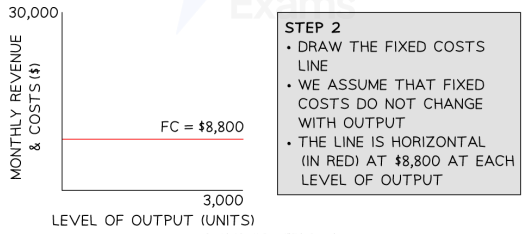
Costs and revenue summary for Tee-Crazy Ltd.

	0 units	500 units	3000 units
Fixed Costs	8,800	8,800	8,800
Variable Costs	0	2,000	12,000
Total Costs	8,800	10,800	20,800
Total Revenue	0	5,000	30,000

Step 2: Draw the axes and plot the fixed costs line

- The **x-axis** should extend to the maximum output level
  - In this case it extends to 3,000 units
- The y-axis should extend to the highest value in the table
  - In this case the highest value is $30,000





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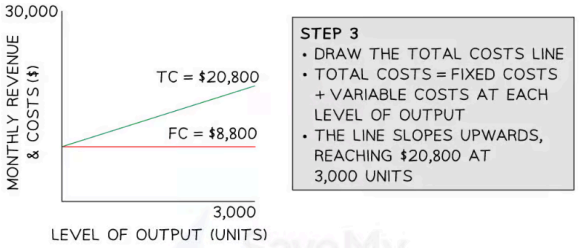
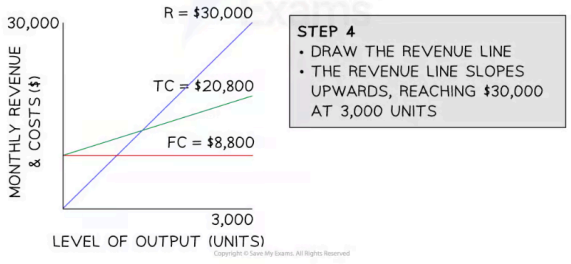
Step 3: Plot the total costs line



- The total cost line slopes upwards, starting at $8,800 at 0 units of output, rising to $20,800 at 3,000 units of output

Step 4: Plot the revenue line



- The revenue line slopes upwards, starting at $0 at 0 units of output, rising to $30,000 at 3,000 units of output

Step 5: Identify the break-even point

- The break-even point is where the revenue line crosses the total costs line
  - Read the break-even output on the x-axis
  - In this case, the break-even point is 1,467 units

Examiner Tips and Tricks

When drawing break-even charts, label all key elements clearly – fixed costs, total costs, total revenue, and break-even point – as missing labels often loses easy marks

output

- The break-even point can be **calculated using the formula**

$$\text{Breakeven point} = \frac{\text{Fixed costs}}{(\text{Selling price} - \text{Variable cost per unit})}$$

Worked Example

Mięsisty Burgers has the following financial information for the month of May

	€ May
Raw materials for each burger	2.10
Packaging for each burger	0.20
Fixed costs	1730
Selling price of each burger	4.95

Using the information in the table, **calculate** the level of output required to break even in May. You are advised to show your workings.

(3)

Step 1: Calculate the variable costs per burger

$$\text{Variable cost per burger} = \text{Raw materials} + \text{Packaging}$$

$$= €2.10 + €0.20 \quad \textbf{(1)}$$

$$= €2.30$$

Step 2: Substitute the values into the break-even formula

$$\text{Break even point in units} = \frac{\text{Fixed Cost}}{(\text{Selling price} - \text{Variable cost})}$$

$$= \frac{1{,}730}{(4.95 - 2.30)} \quad \textbf{(1)}$$

$$= \frac{1{,}730}{(2.65)}$$

$$= 652.83$$

Step 3: Round to the nearest unit

- 653 burgers need to be sold to break even in May **(1)**

Examiner Tips and Tricks

The margin of safety

- **The margin of safety is the amount by which the number of units sold is greater than the break-even point**
  - The margin of safety provides useful information to a firm on how many sales they could lose **before they start making a loss**
- The margin of safety can be calculated using the following formula:

$$\text{Margin of safety} = \text{Quantity of sales} - \text{Break-even level of sales}$$

- Businesses want their margin of safety to be as **large as possible**
  - This means that if demand for their products drops unexpectedly, the business will continue to make a profit

Worked Example

The diagram shows the weekly break-even diagram for the Yorkshire Rare Breed Sausage Company. On average, the business sells 3,200 units each week.

Output, costs and revenues of the Yorkshire Rare Breed Sausage Company

Using the diagram, **calculate** the weekly margin of safety. Show your workings and the formula used.

(2)

Step 1: Identify the break-even output from the graph

= 2,500 units (1)

Step 2: Subtract the break-even output from the actual output

Examiner Tips and Tricks

Use a ruler to help you to read break-even charts accurately.

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